

Indian Institute of Technology Kharagpur

Course : Physics of Waves (PH11003)

Session: 2026–27 (Autumn)

Tutorial 1

1. Using the exponential representations for $\sin \theta$ and $\cos \theta$, verify the following trigonometric identities:

(a) $\sin^2 \theta + \cos^2 \theta = 1$ (b) $\cos^2 \theta - \sin^2 \theta = \cos 2\theta$ (c) $2 \sin \theta \cos \theta = \sin 2\theta$

2. A mass m hangs from a uniform spring of spring constant k .

- (a) What is the period of oscillations in the system?
(b) What would it be if the mass m were hung so that
(1) It was attached to two identical springs hanging side by side?
(2) It was attached to the lower of two identical springs connected end to end?

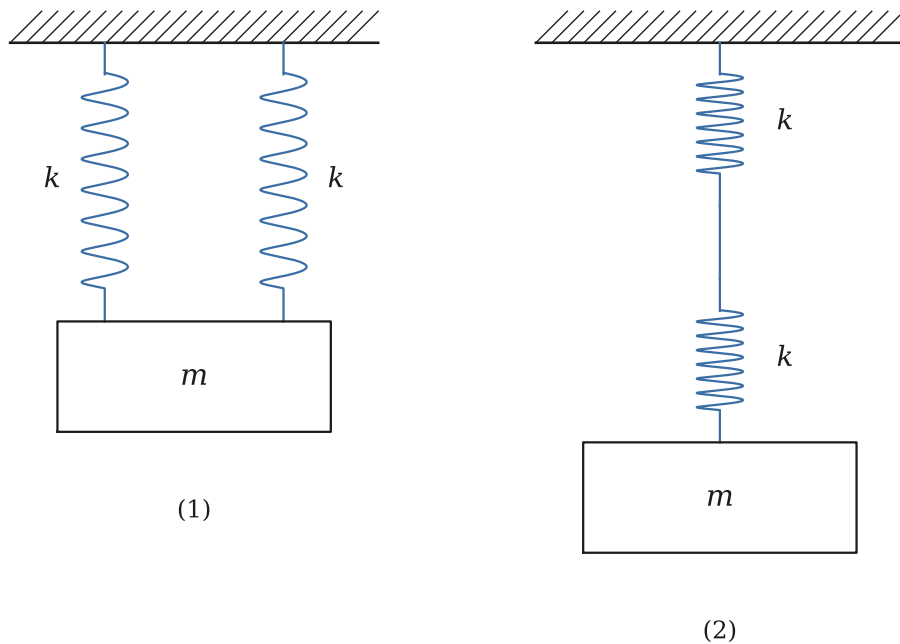


Figure 1: Spring configurations used in Question 2.

3. A cylinder of diameter d floats with l of its length submerged. The total height is L . Assume no damping. At time $t = 0$, the cylinder is pushed down a distance B and released.
- (a) What is the frequency of oscillation?
 - (b) Draw a graph of velocity versus time from $t = 0$ to $t = \text{one period}$. The correct amplitude and phase should be included.
4. Calculate the time average $\langle x^4 \rangle$ for the SHO, where $x = A \cos \omega t$.
5. A uniform rod of mass M and length L is pivoted about one end and oscillates in a vertical plane.
- (a) Find the period of oscillation if the amplitude of the motion is small.
 - (b) Find the average kinetic and potential energy of oscillation.